HYUNWOO KIM

Seoul, South Korea | kimhw7537@gmail.com | https://github.com/Shawn-Hyunwoo

RESEARCH INTEREST

Graduated in Physics and Artificial Intelligence with a primary research focus on practical and experimentally realizable quantum algorithms in the NISQ era. Particularly interested in hardware-efficient quantum machine learning, noise-resilient variational quantum circuits (VQCs), and train-on-classical, deploy-on-quantum approaches that reduce quantum resource requirements. Currently focused on Pauli Propagation–based methods for efficient quantum algorithm design and implementation, aiming to bridge theoretical quantum information science with practical real-world applications.

EDUCATION

BACHELOR OF SCIENCE: PHYSICS & ARTIFICIAL INTELLIGENCE
Sogang University 03/2020 to 02/2026; *Seoul*

- **GPA**: 3.81/4.30 (95.8 percentile; cohort average: 91.4)
- **Honors: Magna Cum Laude**
- **Relevant Physics Coursework**: Quantum Physics, Modern Optics, Statistical Physics, Electromagnetics, Solid-State Physics, Experimental Physics, Classical Dynamics, Thermodynamics, Computational Physics II
  - External coursework: Quantum Information Science, Quantum Computing at Yonsei University via credit exchange
- **Relevant AI & CS Coursework**: Data Structures, Algorithms, Statistics, Linear Algebra, Introduction to Machine Learning, Introduction to Deep Learning, Computational Thinking (Python), Introduction to C Language

EXPERIENCE

NORMA, Inc.
Researcher, Quantum AI Team 07/2025 to Present; *Seoul*

- **Implemented quantum generative models** (QGAN, IQP-QCBM) using Qiskit, PennyLane and NVIDIA CUDA-Q to study expressive power and trainability of variational quantum circuits
- **Developed a Python toolkit for Pauli Propagation–based quantum algorithm implementation**, enabling efficient simulation of operator evolution and implementation of diverse quantum algorithms, including QAOA and VQE
- **Analyzed noise resilience of variational quantum circuits (VQCs),** including VQE and QAOA, under simulation and real-hardware environments
- **Developed a neural-network-based QEM (Quantum Error Mitigation) model** to suppress noise effects in small scale variational quantum circuits

DEPARTMENT OF PHYSCIS, POHANG UNIVERSITY OF SCIENCE AND TECHNOLOGY
Research Assistant (Supervisor: Prof. Yoonho Kim) 07/2024 to 07/2024; *Pohang*

- Research Intern at Quantum Optics and Quantum Information Group (Prof. Yoonho Kim)
- **Designed and aligned a high-precision interferometer and conducted Bell test measurements** using a 1550 nm Kalmár laser for quantum optics research
- **Performed SPDC (Spontaneous Parametric Down-Conversion) experiments** for photon-pair generation and entanglement studies
- **Supported time-energy entanglement experiments**, contributing to data collection and experimental setup optimization

DEPARTMENT OF PHYSICS, SOGANG UNIVERSITY
Research Assistant (Supervisor: Prof. Minsoo Kim) 09/2024 to 12/2024; *Seoul*

- **Analyzed quantum Hall transport data** in mesoscopic 2D systems
- **Studied valley-dependent transport and Dirac-cone** related phenomena
- **Supported theoretical modeling** linking experimental observables to band structure

PUBLICATION & PRESENTATION

Two Approaches of Potential Quantum Advantage with Quantum Generative Models in Drug Discovery

Poster Presentations

- 1st National Undergraduate Quantum Conference — Feb 28, 2026; *Seoul*
- Annual Meeting of the Quantum Information Society of Korea — Feb 25-27, 2026; *Seoul*

Tensorized Pauli Propagation

Poster Presentation

- Conference on Quantum Information, Quantum Korea 2026 — Jul 01-03, 2026; *Seoul*

SAFE ma-QAOA: Surrogate-Assisted and Fine-Tuning Enhanced Multi-Angle QAOA with Parameter Distillation

Preprint

- arXiv: 2605.23377
- Submitted to IEEE International Conference on Quantum Computing and Engineering (QCE) 2026
  Sep 13-18, 2026; *Toronto*

Poster Presentation

- Quantum Future Korea 2026 (2nd National Undergraduate Quantum Conference) — May 15, 2026; Seoul

SELECTED PROJECTS

Realizing Quantum Advantage in the Generation of Drug Library by Quantum Machine Learning

- Conducted research on quantum generative models for molecular drug library generation, integrating **KRAS G12D-binding ligand candidates** into the evaluation pipeline
- A 2024-2026 National Research Foundation of Korea Quantum Computing Challenge Project

Solving Combinatorial Optimization Problem using Quantum Algorithm

- Implemented **QAOA-based solutions for the NP-hard fractional Knapsack Problem** on quantum computers
- Formulated optimization problems as Ising Hamiltonians and applied heuristic optimization strategies

Full-stack Remote Control System for Laboratory Instrumentation

- Developed a **Python-based remote-control system** for digital oscilloscopes to automate experimental data acquisition
- Implemented end-to-end functionality including **device control, data extraction, real-time visualization and analysis** using Python

HONORS & AWARDS

Full Tuition Scholarship (Sogang University, 2020-2025)

- Awarded a merit-based scholarship covering four years of full tuition, granted based on academic performance and GPA criteria

Merit-Based Scholarship

- **Sogang Alumni Association Scholarship** (2024): Stipend-based scholarship awarded through academic excellence and personal statement evaluation
- **Catholic Outstanding Student Scholarship** (2025): Stipend-based scholarship awarded based on academic achievement and personal evaluation

Residential Scholarship

- **Accommodation Scholarship** (Lotte Scholarship Foundation, 08/2023 – 07/2025): Two-year fully funded accommodation awarded through academic and personal evaluation

TRAINING

Seoul National University — 07/2024 to 08/2025

- The Basics of Quantum Computing and Information

LG AI Institute 01/2024 to 02/2024

- Acquired expertise in advanced machine learning techniques
- Developed AI model that predicts successful sales conversion using Marketing Qualified Lead (MQL) customer information

TECHNICAL SKILLS

- **Experimental Physics:** Interferometry, SPDC, Superconductivity of graphene, Hall effect
- **Quantum SDKs**: Qiskit, PennyLane, NVIDIA CUDA-Q
- **Programming**: Python (NumPy, SciPy, PyTorch), C

TEACHING

INSTITUTE FOR CONVERGENCE EDUCATION, SOGANG UNIVERSITY
Teaching Assistant 03/2024 to 06/2025; *Seoul*

- Lecture: Introduction to Artificial Intelligence Programming (Python)
- Instructed undergraduate students on core Python programming principles and problem-solving techniques
- Coordinated with the professor and other TAs to prepare course materials, effectively addressing student inquiries, debugging code, and assisting with practical exercises

ADDITIONAL EXPERIENCE

MILITARY POLICE, REPUBLIC OF KOREA, ARMY
SDT (Special Duty Team) Sniper 02/2021 to 08/2022; *Gangwon-do*

- Served as a sniper for SDT (Special Duty Team) responsible for counterterrorism and commander's security
- Managed and calibrated precision optical equipment and weaponry for optimal operational readiness